

CUT BANK, MT, USA

WMO: 727690

Lat: 48.603N

Lon: 112.375W

Elev: 3838

StdP: 12.77

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24137

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-12.4	-24.4	1.6	-16.9	-18.3	2.3	-11.2	48.0	40.5	42.3	37.2	5.8	340	0.779

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	89.5	59.8	85.8	58.8	82.1	57.9	62.4	81.3	60.8	80.1	59.4	78.3	12.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.7	76.2	66.5	54.0	71.5	65.5	52.0	66.3	64.4	29.8	80.9	28.6	80.2	27.5	78.6	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.2	33.3	29.0	DB	-28.1	94.9	7.4	3.5	-33.4	97.4	-37.7	99.5	-41.9	101.5	-47.3	104.0	(4)
(5)			WB	-26.2	65.8	5.3	2.1	-30.0	67.2	-33.1	68.5	-36.1	69.6	-40.0	71.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	22.3	22.7	31.0	40.2	49.5	57.4	65.7	63.9	54.8	42.4	31.0	22.4		
	DBStd	19.16	16.01	15.63	13.39	9.04	7.96	6.21	5.92	6.28	8.44	10.08	13.76	14.84		
	HDD50	4391	859	765	591	308	107	10	1	2	46	270	576	856		
	HDD65	8543	1323	1184	1054	744	481	238	63	95	317	702	1021	1321		
	CDD50	1491	1	0	2	15	92	233	487	434	189	33	5	0		
	CDD65	169	0	0	0	0	1	11	84	63	10	0	0	0		
	CDH74	3524	0	0	0	3	75	261	1551	1275	350	9	0	0		
	CDH80	1221	0	0	0	0	9	59	596	456	101	0	0	0		
	WSAvg	12.5	14.4	12.8	13.2	13.0	12.2	11.9	10.3	9.9	10.8	13.0	14.2	14.1		
	PrecAvg	11.7	0.4	0.2	0.5	0.9	2.0	2.6	1.4	1.5	1.1	0.4	0.4	0.3		
	PrecMax	23.3	1.3	0.7	1.8	2.3	6.4	8.0	9.0	3.8	4.1	2.2	1.3	1.3		
(15) Precipitation	PrecMin	5.0	0.0	0.0	0.0	0.0	0.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	4.2	0.3	0.2	0.4	0.6	1.2	2.0	1.6	1.1	1.0	0.4	0.3	0.3		
	0.4%	DB	53.9	55.7	62.8	72.7	80.8	86.7	94.2	92.5	88.1	74.6	63.8	52.5		
	MCWB	40.7	40.9	45.1	50.1	56.0	61.2	61.1	59.9	58.3	52.4	47.0	40.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	48.5	50.0	57.1	66.5	75.4	80.9	90.3	88.6	82.6	68.0	56.8	47.6		
	MCWB	38.5	38.9	42.7	47.7	54.1	58.3	60.4	59.0	55.8	49.2	43.8	37.7			
	5%	DB	45.2	45.8	53.3	61.3	71.4	76.5	86.6	85.3	78.1	62.7	52.4	44.6		
	MCWB	36.7	36.3	40.9	45.1	52.3	56.7	59.6	58.3	54.7	47.6	41.1	36.1			
	10%	DB	41.9	42.4	48.7	55.9	66.0	72.4	82.5	81.7	72.6	57.3	48.3	40.7		
	MCWB	34.8	34.5	38.4	42.5	49.8	54.9	58.8	57.3	52.9	44.6	39.3	33.8			
	0.4%	WB	42.8	42.9	46.8	51.9	57.3	63.7	65.5	63.1	60.3	54.1	48.8	42.1		
	MCDB	51.1	53.0	59.8	69.4	76.2	80.6	83.2	82.9	79.9	70.7	61.4	50.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.4	39.5	43.7	48.5	55.0	60.4	63.2	61.2	57.7	50.7	44.7	38.8		
	MCDB	47.5	48.9	55.4	65.2	72.9	76.3	81.5	81.9	78.2	65.7	55.6	46.5			
	5%	WB	37.1	36.9	41.2	45.9	53.0	58.2	61.6	59.7	55.4	48.1	41.9	36.3		
	MCDB	44.6	45.2	52.1	60.2	68.7	73.0	81.0	80.2	75.1	61.0	51.0	43.7			
	10%	WB	34.9	34.6	38.8	43.3	50.8	56.2	60.1	58.4	53.4	45.6	39.4	33.9		
	MCDB	41.6	42.1	48.2	55.0	64.1	69.8	79.2	78.3	71.7	57.1	47.7	40.5			
(35) Mean Daily Temperature Range	MDBR	18.6	19.7	20.6	22.1	23.1	23.5	28.9	29.1	26.3	22.3	18.6	18.0			
	5% DB	MCDBR	19.9	23.2	26.8	31.4	32.1	31.3	35.7	36.7	36.7	29.8	22.2	18.8		
	MCWBR	13.4	14.8	15.8	16.5	15.1	13.5	12.8	14.0	15.8	15.9	12.9	12.5			
	5% WB	MCDBR	19.3	22.6	24.9	29.5	29.0	28.0	31.3	31.4	33.4	28.1	21.7	18.3		
	MCWBR	13.8	15.0	15.2	16.0	14.0	13.2	12.5	12.7	15.0	15.6	13.4	12.7			
(40) Clear-Sky Solar Irradiance	taub	0.246	0.255	0.275	0.309	0.320	0.322	0.324	0.341	0.307	0.278	0.253	0.236			
	taud	2.465	2.484	2.494	2.422	2.442	2.486	2.487	2.410	2.478	2.544	2.519	2.463			
	Ebn at Noon	271	294	303	299	298	298	295	285	286	280	266	261			
	Edh at Noon	19	24	28	33	34	33	33	34	28	22	18	17			
(44) All-Sky Solar Radiation	RadAvg	393	706	1085	1492	1780	1998	2268	1838	1308	795	436	314			
	RadStd	31	55	102	92	159	161	139	124	106	74	31	26			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page